

Homework 4

5.7. Show that the quantum-mechanical partition function of a system of N interacting particles approaches the classical form

$$Q_N(V, T) = \frac{1}{N! h^{3N}} \int e^{-\beta E(q, p)} d^{3N} q d^{3N} p$$

as the mean thermal wavelength λ becomes much smaller than (i) the mean interparticle distance $(V/N)^{1/3}$ and (ii) a characteristic length r_0 of the interparticle potential.⁹

Goal : Show that Q_N has the form above for N interacting quantum particles

Solution

In quantum mechanical systems,

$$Q_N = \text{Tr} (e^{-\beta \hat{H}})$$

← distinguishable → identical

$$Q_N = \int d^{3N} \vec{r} \langle \vec{r} | e^{-\beta \hat{H}} | \vec{r} \rangle$$

$$Q_N = \int d^{3N} \vec{r} \langle \vec{r} | \hat{\Pi}_{\pm} e^{-\beta \hat{H}} | \vec{r} \rangle$$

where $\hat{\Pi}_{\pm}$ is the symmetrization projector such that

$$\hat{\Pi}_{\pm} = \frac{1}{N!} \sum_P \delta_P \hat{P}$$

← avoid over counting permutations

← Summing over all possible unique permutations

$\delta_P = 1$ (bosons)

$\delta_P = \text{sgn}(P)$ fermions

← Permutation operator

$$\therefore Q_N = \frac{1}{N!} \sum_P \delta_P \int d^{3N} \vec{r} \langle \vec{r} | \hat{P} e^{-\beta \hat{H}} | \vec{r} \rangle$$

The hamiltonian $\hat{H}$ is invariant under permutations $\hat{P}$, $\therefore \hat{P}$ commutes with $\hat{H}$ i.e. $\langle \vec{r} | \hat{P} e^{-\beta \hat{H}} | \vec{r} \rangle = \langle \hat{P} \vec{r} | e^{-\beta \hat{H}} | \vec{r} \rangle$

We are at the finish line. We now assess some approximations we can make!

① Since $\lambda \ll (V/N)^{1/3}$, wave functions overlaps are rare.

Therefore, only the Identity permutation contributes i.e.

$$\sum_P \delta_P P = I$$

② Using Trotter formula,

$$e^{-\beta \hat{H}} = e^{-\beta \hat{K}} e^{-\beta \hat{V}} \approx e^{-\beta \hat{V}/2} e^{-\beta \hat{K}} e^{-\beta \hat{V}/2}$$

$$\langle \vec{r} | e^{-\beta \hat{H}} | \vec{r}' \rangle \approx e^{-\beta \hat{V}(\vec{r}')/2} \langle \vec{r} | e^{-\beta \hat{K}} | \vec{r}' \rangle e^{-\beta \hat{V}(\vec{r})/2}$$

$$\approx e^{-\beta \hat{V}(\frac{\vec{r} + \vec{r}'}{2})} \langle \vec{r} | e^{-\beta \hat{K}} | \vec{r}' \rangle$$

$$\approx e^{-\beta \hat{V}(\frac{\vec{r} + \vec{r}'}{2})} \cdot \frac{1}{\lambda^{3N}} \exp \left\{ \frac{-\pi |\vec{r} - \vec{r}'|^2}{\lambda^2} \right\}$$

(5.5.18)

$$\therefore Q_N \approx \frac{1}{N!} (I) \int d^{3N} \vec{r} \langle \vec{r} | e^{-\beta \hat{H}} | \vec{r} \rangle$$

$$= \frac{1}{N!} \frac{1}{\lambda^{3N}} \int d^{3N} \vec{r} e^{-\beta V(\vec{r})}$$

Introducing the integral over momentum states

$$\int d^3 p e^{-\beta p^2/2m} = \left(\frac{2\pi m}{\beta} \right)^{3/2}$$

$$\int d^{3N} p e^{-\beta \sum_i^N p_i^2/2m} = \left(\frac{2\pi m}{\beta} \right)^{3N/2} = \frac{h^{3N}}{\lambda^{3N}}$$

$$Q_N \approx \frac{1}{N!} \frac{1}{\lambda^{3N}} \int d^{3N} \vec{r} e^{-\beta V(\vec{r})} \cdot \frac{\lambda^{3N}}{h^{3N}} \int d^{3N} \vec{p} e^{-\sum_i^N \frac{p_i^2}{2m}}$$

$$\approx \frac{1}{N!} \int d^{3N} \vec{r} d^{3N} \vec{p} e^{-\beta \hat{H}}$$



5.18. Use particle in a box wavefunctions to construct the N -body quantum canonical partition function $Q_N(V, T)$ for indistinguishable noninteracting particles with mass m in a rectangular box with hard walls and volume $V = L_x L_y L_z$. Use the delabeling factor $1/N!$ to account for indistinguishability. Show that the partition function factorizes: $Q_N(V, T) = Q_1^N(V, T)/N!$. Show

Goal : Show that $Q_N(V, T) = Q_1^N(V, T) / N!$

Solution

We will find Q_1 & Q_N and show that they have the appropriate relation.

The solution to the wavefunction of a particle of mass m in 3D box :

$$0 < x < L_x, \quad 0 < y < L_y, \quad 0 < z < L_z$$

$$\psi_{n_x, n_y, n_z} \propto \sin(k_x x) \sin(k_y y) \sin(k_z z) \quad \text{where } k_i = \frac{n_i \pi}{L_i} \\ \text{for } i \in (x, y, z) \\ \& n_i \in \mathbb{N}$$

The energy eigenvalues $\epsilon_{n_x n_y n_z}$:

$$\epsilon_{n_x n_y n_z} = \frac{\hbar^2}{2m} \sum_i k_i^2 = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right)$$

The partition function for one particle :

$$Q_1(V, T) = \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} e^{-\beta \epsilon_{n_x n_y n_z}}$$

Let $\alpha_i = (n_x^{(i)}, n_y^{(i)}, n_z^{(i)})$ for each particle i from $1, 2, \dots, N$

$$Q_N(V, T) = \underbrace{\frac{1}{N!}}_{\text{indistinguishable}} \sum_{\alpha_1} \sum_{\alpha_2} \dots \sum_{\alpha_N} e^{-\beta E_{\alpha_1, \alpha_2, \dots, \alpha_N}}$$

Since particles are non-interacting particles &

1. Hamiltonian is purely kinetic.
2. Hamiltonian of N-body particles is the sum of the single particle Hamiltonian

$$E_{\alpha_1, \alpha_2, \dots, \alpha_N} = \sum_{j=1}^N \epsilon_{\alpha_j} \quad \text{with } \epsilon_j \text{ energy of the } j^{\text{th}} \text{ particle}$$

We return to $Q_N(V, T)$

$$Q_N(V, T) = \frac{1}{N!} \sum_{\alpha_1} \sum_{\alpha_2} \dots \sum_{\alpha_N} e^{-\beta E_{\alpha_1, \alpha_2, \dots, \alpha_N}}$$

$$= \frac{1}{N!} \sum_{\alpha_1} \sum_{\alpha_2} \dots \sum_{\alpha_N} e^{-\beta \sum_{j=1}^N \epsilon_{\alpha_j}}$$

$$= \frac{1}{N!} \sum_{\alpha_1} \sum_{\alpha_2} \dots \sum_{\alpha_N} \prod_{j=1}^N e^{-\beta \epsilon_{\alpha_j}}$$

$$= \frac{1}{N!} \sum_{\alpha_1} e^{-\beta \epsilon_{\alpha_1}} \sum_{\alpha_2} e^{-\beta \epsilon_{\alpha_2}} \dots \sum_{\alpha_N} e^{-\beta \epsilon_{\alpha_N}}$$

$$= \frac{1}{N!} \left(\sum_{\alpha_i} e^{-\beta \epsilon_{\alpha_i}} \right)^N$$

$$= \frac{1}{N!} [Q_1(V, T)]^N \quad \square$$

6.10. (a) Show that, if the temperature is uniform, the pressure of a classical gas in a uniform gravitational field decreases with height according to the *barometric formula*

$$P(z) = P(0) \exp \{-mgz/kT\},$$

where the various symbols have their usual meanings.¹⁷

Goal: Find $P(z)$ for a classical gas

Solution

Investigating the forces of a classical gas in a volume dV .

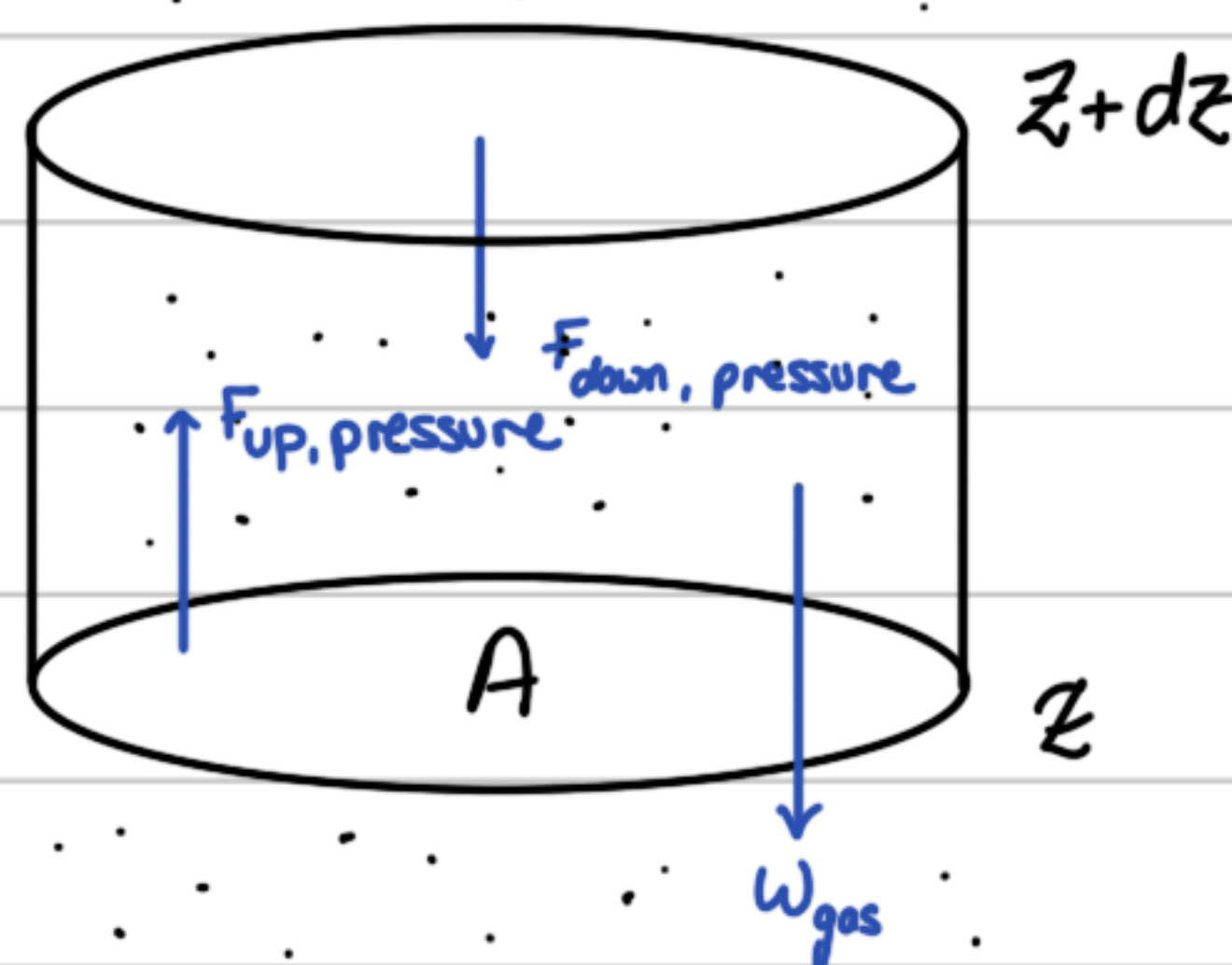
In equilibrium,

$$\sum F = 0$$

$$F_{\text{up, pressure}} + F_{\text{down, pressure}} + W_{\text{gas}} = 0$$

$$P(z)A - P(z+dz)A - mg = 0$$

pressure
cross-sectional area
mass of gas in dV
gravitational acc.



For mass m , $m = \rho dV = \rho A dz$

$$-P(z+dz)A + P(z)A - \rho A dz g = 0$$

$$\frac{P(z+dz) - P(z)}{dz} = -\rho g$$

For a classical gas: $P = nkT$, $\rho = nm$

pressure
number of particles
mass of a particle

$$\therefore \rho = \frac{P}{kT} m$$

Going back to

$$\frac{P(z+dz) - P(z)}{dz} \approx \frac{dP}{dz} \approx -\frac{P}{kT} mg$$

$$\int \frac{1}{P} dP = \int -\frac{mg}{kT} dz$$

$$\ln P = -\frac{mg}{kT} z + C$$

$$P(z) = P_0 \exp\left(-\frac{mgz}{kT}\right) \quad \square$$

7.20. The (canonical) partition function of the blackbody radiation may be written as

$$Q(V, T) = \prod_{\omega} Q_1(\omega, T),$$

so that

$$\ln Q(V, T) = \sum_{\omega} \ln Q_1(\omega, T) \approx \int_0^{\infty} \ln Q_1(\omega, T) g(\omega) d\omega;$$

here, $Q_1(\omega, T)$ is the single-oscillator partition function given by equation (3.8.14) and $g(\omega)$ is the density of states given by equation (7.3.2). Using this information, evaluate the Helmholtz free energy of the system and derive other thermodynamic properties such as the pressure P and the (thermal) energy density U/V . Compare your results with the ones derived in Section 7.3 from the q -potential of the system.

Goal : ① Find the Helmholtz free energy $A(V, T)$
② Find pressure P & thermal density U/V .

Solution

Single - Oscillator partition function :

$$Q_1(V, T) = \frac{e^{-\frac{1}{2}\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \quad (3.8.14)$$

$$\ln Q_1 = -\frac{1}{2}\beta\hbar\omega - \ln(1 - e^{-\beta\hbar\omega})$$

$$A_1 = -kT \ln Q_1 = \frac{\hbar\omega}{2} + kT \ln(1 - e^{-\beta\hbar\omega})$$

However, we would like to find A for black body radiation. We only consider the thermal part alone ignoring $\frac{\hbar\omega}{2}$.

$$\begin{aligned} A(V, T) &= -kT \ln Q \approx \int_0^{\infty} \ln Q_1 g(\omega) d\omega \quad (\text{question}) \\ &= \int_0^{\infty} kT \ln(1 - e^{-\beta\hbar\omega}) \frac{\omega^2 V}{\pi^2 c^3} d\omega \quad (7.3.2) \end{aligned}$$

$$\begin{aligned}
A(V, T) &\approx \int_0^\infty kT \ln(1 - e^{-\beta \hbar \omega}) \frac{\omega^2 V}{\pi^2 c^3} d\omega \\
&= \frac{V k T}{\pi^2 c^3} \int_0^\infty \underbrace{\ln(1 - e^{-\beta \hbar \omega})}_u \underbrace{\omega^2 d\omega}_{dv} \\
&= \frac{V k T}{\pi^2 c^3} \left[\cancel{\ln(1 - e^{-\beta \hbar \omega})} \frac{\omega^3}{3} \Big|_0^\infty - \int_0^\infty \frac{-(-1)\beta \hbar e^{-\beta \hbar \omega}}{1 - e^{-\beta \hbar \omega}} \frac{\omega^3}{3} d\omega \right] \\
&\quad \text{0 by graph} \\
&= -\frac{V \hbar}{3 \pi^2 c^3} \int_0^\infty \frac{\omega^3}{e^{\beta \hbar \omega} - 1} d\omega \quad \begin{array}{l} x = \beta \hbar \omega \\ dx = \beta \hbar d\omega \end{array} \\
&= -\frac{V \hbar}{3 \pi^2 c^3} \cdot \frac{1}{(\beta \hbar)^3} \cdot \frac{1}{\beta \hbar} \underbrace{\int_0^\infty \frac{x^3}{e^x - 1} dx}_I
\end{aligned}$$

For integral I ,

$$\begin{aligned}
\frac{1}{e^x - 1} &= \frac{e^{-x}}{1 - e^{-x}} = e^{-x} (1 + e^{-x} + (e^{-x})^2 + \dots) \\
&= \sum_{n=1}^{\infty} (e^{-x})^n
\end{aligned}$$

$$\begin{aligned}
I &= \int_0^\infty \frac{x^3}{e^x - 1} dx = \int_0^\infty x^3 \sum_{n=1}^{\infty} e^{-nx} dx \\
&= \sum_{n=1}^{\infty} \int_0^\infty x^3 e^{-nx} dx \\
&= \sum_{n=1}^{\infty} \frac{3!}{n^4} = 6 \sum_{n=1}^{\infty} \frac{1}{n^4} = 6 \zeta(4) \\
&= 6 \frac{\pi^4}{90} = \frac{\pi^4}{15}
\end{aligned}$$

Back to $A(V, T)$

$$A(V, T) = -\frac{V}{3} \frac{(kT)^4}{\pi^2 c^3 \hbar^3} \cdot \frac{\pi^4}{15} = -\frac{\pi^2 V k^4 T^4}{45 \hbar^3 c^3}$$

Thermodynamic Quantities 8

$$S = \left(- \frac{\partial A}{\partial T} \right)_V = - \frac{4A}{T}$$

$$P = \left(- \frac{\partial A}{\partial V} \right)_T = - \frac{A}{V}$$

$$U = A + ST = -3A = 3PV \quad \Rightarrow \quad \frac{U}{V} = 3P$$

$$\text{OR} \quad P = \frac{1}{3} \frac{U}{V}$$